

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Montana Nicholas	Student Number:	C00297005
Working Title:	RATS!		
Description:	RATS! Is an original strategic tabletop game set inside a house that has become a battlefield between a colony of rats and a household cat or cats. Players take on the role of either the rats, who must navigate through the house and ultimately reach the fridge in the kitchen, or the cats, who must prevent them from doing so. The game will feature a tile based board representing different areas in the house, including the kitchen, living room, bedroom, bathroom and connecting hallways. Rats will be able to use hidden routes such as cracks, vents, and pipes to navigate the house, while cats can move throughout the boards, place traps, and pursue the rats. Players will use dice for movement, while cards will provide different abilities, attacks, healing and other gameplay actions. Combat between cats and rats will incorporate card based mechanics and dice based probability. These gameplay systems will be developed, prototyped and refined through research and playtesting throughout the project The physical game will be the primary outcome of the project. This will involve designing the games visual identities, characters, environments, board, cards, tokens, traps and other components through 2D concept art, 3D modelling, and 3D printing. The project will also include the design and production of packaging, and documentation for the completed game. Alongside the physical game, an interactive digital companion will be developed using Svelte. Rather than functioning as a separate digital version of the game, the companion will be designed to support the physical game experience. It will provide access to the game rules, character and card information, game setup guidance and other supporting information. Additional features such as a digital dice roller, health tracker, card drawing tool and interactive 3D representations of game assets or the house environment will be considered during development. Three.js may also be explored as a method of displaying and interacting with 3D assets within the web application. The digital component will be developed alongside the physical game, with its features being decided and refined as the physical game is developed and tested.		
Reasons for selecting project:	I am interested in developing this project because it combines several areas of Interactive Digital Art and Design that I enjoy,		

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	particularly character design, 3D modelling, visual design and physical prototyping. I am interested in creating a physical product rather than only a digital experience, and I would like the final outcome to feel like a complete professionally designed game rather than a collection of individual assets. This project will allow me to develop my skills in areas that I have previously only worked with briefly at an introductory level, particularly in 3D modelling, 3D printing and web development. I am also really interested in the process of designing a game from the initial concept through to a playable physical prototype. This will allow me to experiment with gameplay mechanics, test the game with players and make changes based on feedback through the development process. The project will also allow me to explore how physical and digital media can work together for one interactive experience. The physical board game will be supported by an interactive web application, allowing me to incorporate programming and UI/UX design into the project without making the digital application the primary outcome.
Proposed research content:	<u>Tabletop Game Design</u> - Research will focus on the design and development of tabletop games, particularly asymmetric games, where different players have different objectives and abilities. <u>Board Game Mechanics</u> These games will be researched mainly for their gameplay systems, including movement, combat, territory control, asymmetric abilities, resource management, cards, dice and player interaction - Risk: territory control, movement and strategic decision making - Cluedo: hidden information and exploration - Root: asymmetric factions, different abilities and objectives and territory control - King of Tokyo: dice-based gameplay, health, and combat <u>Visual Design</u> These games will be researched for their visual identity, character design, illustration style, colour and presentation. These references are mainly relevant to the playful, cartoon like visual style I would like to develop for RATS! - My little scythe: colourful environments, simplified

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	character designs and a playful interpretation of strategic gameplay • Exploding kittens: simple character illustrations, humour and card design • Sushi Go!: simple illustrations, colour coding and clear card layouts • Mice & Mystics: storybook style character design and representation of a large environment from the perspective of small characters <u>Digital Companion and Web Development</u> I will research how digital technology can be used to support a physical tabletop game without replacing the physical experience. Existing physical/digital hybrid board games will be examined to explore how digital tools can be used for game setup, rules, information, randomisation and other supporting functions. I will also research interactive web design and UI/UX, including navigation, information architecture, responsive design and presenting game information clearly. Svelte will be used as the main web framework for developing the digital companion, while Three.js may be explored for displaying and interacting with 3D game assets within the browser <u>3D Modelling and Physical Production</u> My research will cover the use of 3D modelling and 3D printing in the development of tabletop game components. This will include consideration of scale, usability, durability, printability and how different components can be designed for practical use during gameplay. The research will inform the development of the game's character pieces, environmental objects, traps and other physical components. <u>Playtesting and Development</u> Playtesting will be used throughout the development of RATS! to identify problems with the game's mechanics, balance, usability and overall player experience. Gameplay sessions will be used to test areas such as movement, combat, cards, traps and the different objectives of the cats and rats. Feedback from playtesting will then be used to refine the game before producing the final physical prototype.
External links (if applicable):	Game Design / Board Game Research • A data driven review of board game design and

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	interactions of their mechanics, Samarasinghe et al. (2021) Link • Don't Flip the Table Yet: A Framework for the Analysis of Visual and Cognitive Ergonomics in Board Games - Passarelli et al. (2024) Link • Meeple Centred Design: A Heuristic Toolkit for Evaluating the Accessibility of Tabletop Games - Heron et al. (2018) Link Physical / 3D Printing • Current and Future Opportunities for 3D Printing in Modern Board Games - Simões (2023) Link Digital/ Hybrid Games • Hybrid Board Game Design Guidelines - Kankainen & Paavilainen (2019) Link • Examining the Digital in Hybrid Board Games - University of Melbourne Link Technology • Three.js Documentation Link
Hardware requirements:	• Personal computer/laptop to run 3D modelling and development software • 3D printer • Filament/resin and other 3D printing materials • Materials for physical board and prototype construction • Cutting/Assembly tools as required • Tablet, phone, laptop etc to test the digital companion
Software requirements:	• Maya/Blender: for 3D modelling • Svelte, HTML/CSS/JavaScript: web development • Visual Studio Code: programming and development • GitHub • Penpot: UI/UX planning and prototyping • Krita: concept art and any visual assets • Three.js: potentially used for interactive 3D components on the website • 3D printer slicing software?: prepping models for 3D printing
Other requirements:	Play testers: I would need participants for testing and evaluating the physical board game throughout the development

Signed:
MONTANA NICHOLAS

Date:
20/09/2026

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Approved/Not Approved:		
Reasons for not approving project:		
Conditions attached to approving project:		
Approved/Not Approved:		
Name of Supervisor:		
Signed: _____	Date: _____	